Tobias Hullette

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EDUCATION

North Carolina State University **August 2024 – May 2026** ***(Anticipated)***
Master of Science | **Aerospace Engineering** | Park Scholar

North Carolina State University **August 2021 – May 2024**
Bachelor of Science | **Aerospace Engineering** | Park Scholar

EXPERIENCE

REACHR – NASA Research Grant **November 2024 – Present**
P.I. (Principal Investigator)

- Designed, analyzed, built, tested, and flew a multi-mode VTOL-to-Cruise, multi-domain air vehicle for post hurricane surveillance and relief delivery.
- Won NASA USRC grant and received $80,000 to continue R&D.
- Manage a ten-person team of PhD, masters, and undergraduate students in design, analysis (FEA and CFD), manufacturing, and testing of full-scale, mission capable composite Unmanned Aerial Vehicle, HERO 2.
- Leads research in implementing autonomous, multi-mode drone technology in the emergency response industry.

RARE Lab – North Carolina State University (Research) **July 2024 – Present**
Graduate Research Assistant

- Design and build automated experimental setups for force-torque and particle image velocimetry rotorcraft aerodynamics studies.
- Run experiments with Arduino and DEWESoft DAQ and LaVision particle image velocimetry.
- Post-process and analyze data in MATLAB, DaVis11, DeweSoft, and NI.
- Presented research at 2025 Vertical Flight Society – Forum 81 conference.

Northrop Grumman Corporation **Summer 2023 & Summer 2024**
Advanced Manufacturing Systems Engineering - Composites Intern

- Obtained a Department of Defense (DoD) Secret Clearance (Active)
- Designed a COTs-based system to identify composite layup vacuum bag leaks on the factory floor, pushed the solution through testing and implementation approval, saving over $1M annually on a single classified program.
- Designed MATLAB apps to visualize real time heated cure table uniformity for flat panel layups.
- Coded a multi-factor model for oven, heated cure table, and autoclave usage for optimizing batch part production scheduling.
- Coordinated intern tasking to help launch the Northrop Grumman Palmdale Fabrication Laboratory.

ISSRL Lab - North Carolina State University (Research) **October 2021 – April 2023**
Mechanical Design and Fabrication Lead

- Designed, analyzed, fabricated, and assembled parts for the MantaRay Underwater Hydrokinetic Energy Harvesting Vehicle funded by Defense Advanced Research Projects Agency (DARPA) and the DoD.
- Led graduate and PhD researchers in vehicle design, FEA analysis, manufacturing, and assembly tasks.
- Built and maintained bill of materials, coordinated material purchasing with vendors, and managed part inventory.

SKILLS

- 2D/3D Design (CATIA, NX, SolidWorks, OnShape, AutoCAD)
- Computational Analysis (Ansys Structural FEA, Ansys Fluent CFD)
- Programming (MATLAB, Python, C++, G-Code)
- Manufacturing (Metallic, Composites, Assemblies)
- GD&T, Vendor Review for Manufacturability
- Problem Solving and Work Ethic
- Leadership and Teamwork
- Writing and Communication

Tobias Hullette
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September 23, 2025

Dear BETA Technologies Hiring Manager,

I am excited to apply for the Composites and Manufacturing Engineer position on the Advanced Concepts team in South Burlington. As a graduate student in Aerospace Engineering with hands-on experience in composite design, manufacturing, and rapid prototyping, I have been working in exactly the type of "concept-to-flight-test" environment that this role requires. You will find me to be an ambitious problem solver that works well with others and is passionate about sustainable aviation.

Currently, I serve as Principal Investigator on an $80,000 NASA-funded project, REACHR, developing a composite lift-plus-cruise, tilt-prop UAV for natural disaster response. This role requires full-lifecycle ownership: CAD design of detailed assemblies including composites with associated manufacturing models and tooling, coordination with internal and external fabricators to see parts through manufacturing, and integration of different systems within the aircraft structure. I develop and build first article components, assemble prototypes, and run experiments to inform future designs while managing material systems requirements and associated tooling. This experience has deepened my commitment to engineering solutions that serve people and has grown my ability to manage concurrent tasks and lead teams effectively.

The past two summers at Northrop Grumman's advanced composites facility in Palmdale greatly prepared me to build the REACHR project. At Northrop Grumman, I improved composite fabrication and assembly workflows, working directly with suppliers on R&D composite tooling development and supporting layup methodology development for first article components. I developed scalable manufacturing processes in parallel with product design, identifying failure points, developing corrective tooling solutions, and implementing changes that drastically reduced manufacturing time, saving over $1M annually on composite programs.

On the $3M DARPA MantaRay program, I began as a CAD technician and grew into the Mechanical Design and Fabrication Lead position, managing detailed assemblies and creating technical data packages and BOMs for composite components. I developed material systems for demanding sub-sea conditions with emphasis on low-cost prototype pathways, working extensively with external fabricators on the design and layup process for our composite wing. I quickly grew accustomed to long days at the lake, testing prototypes, and long nights in the lab, improving designs, coordinating with manufacturers, assembling new systems, and preparing for the next day of testing.

Throughout my experience, I have participated in cross-functional design reviews, followed rigorous design, review, and change processes, and worked to improve manufacturing processes. I bring technical expertise in composite design and manufacturing, a results-focused mindset, and a collaborative spirit to deliver robust aircraft systems from white paper to operational deployment. I am excited to discuss how I can grow and apply my experience as a BETA engineer, contributing to the revolution in electric aviation.

Sincerely,
Tobias Hullette